Time Response of Feedback Control Systems — Exam Revision Notes

1. Time Response of Control Systems

What it is:
Time response is the variation of system output with time for a given input. It consists of **transient response** and **steady-state response**.

Key Formulas

- Total response = Transient response + Steady-state response
- Transient response:

$$\lim_{t\to\infty} C_t(t) = 0$$

- Steady-state response:

$$C_{ss} = \lim_{t\to\infty} C(t)$$

Template (How to Solve Problems)

1. Identify input signal.
2. Find transfer function $C(s)/R(s)$.
3. Compute output $C(s) = G(s)R(s)$.
4. Take inverse Laplace.
5. Evaluate:
    - transient terms (exponentials)
    - steady state value

Facts / Tricks / Recognition Patterns

- Presence of **energy storage elements** ⇒ transient exists.
- Final Value Theorem used for steady state:

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s)$$

Example (Short)

Find steady state value of $C(s) = \frac{10}{s(s+2)}$
Use Final Value Theorem.

2. Standard Test Signals

What it is:
Standard inputs used to analyze system performance.

laplace

Key Formulas

- Step:

$$r(t) = Au(t), \quad R(s) = \frac{A}{s}$$

- Ramp:

$$r(t) = At, \quad R(s) = \frac{A}{s^2}$$

- Parabolic:

$$r(t) = \frac{A}{2}t^2, \quad R(s) = \frac{A}{s^3}$$

- Impulse:

$$\delta(t), \quad R(s) = 1$$

Template

1. Identify input type.
2. Substitute $R(s)$.
3. Compute $C(s)$.
4. Use inverse Laplace.

Tricks

- Step → position control
- Ramp → velocity systems
- Parabolic → acceleration systems

Example

Ramp input ⇒ use $R(s) = 1/s^2$.

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3. Steady State Error (General Expression)

What it is:
Difference between reference input and output at steady state.

Key Formulas

Error signal:

$$E(s) = \frac{R(s)}{1 + G(s)H(s)}$$

→ for negative feedback

if +ve feedback, $\frac{R(s)}{1 - G(s)H(s)}$.

Steady state error:

$$e_{ss} = \lim_{s \to 0} sE(s)$$

Unity feedback:

$$e(t) = r(t) - c(t)$$

Template

1. Write $E(s) = R(s)/(1 + GH)$
2. Multiply by s
3. Apply limit $s \to 0$

Tricks

- Always apply Final Value Theorem.
- For unity feedback, simplify first.

Common Mistakes

- Forgetting factor s before limit.
- Not checking system stability.

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4. First Order System Response

What it is:
System with denominator order = 1.

Transfer function:

$$G(s) = \frac{1}{Ts+1}$$

Key Formulas

Step input:

$$C(t) = 1 - e^{-t/T}$$

$$e_{ss} = 0$$

Impulse input:

$$C(t) = \frac{1}{T}e^{-t/T}$$

Ramp input:

$$C(t) = t - T(1 - e^{-t/T})$$

$$e_{ss} = T$$

Time constant rule:

$$C(T) = 0.632\,(\text{final value})$$

Template

1. Identify time constant T.
2. Substitute input Laplace.
3. Partial fractions.
4. Inverse Laplace.

Tricks

- At $t = T$ ⇒ 63.2% value.
- Smaller T ⇒ faster response.

Example

Find step response for $G(s) = 1/(5s+1)$.

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5. Second Order System (Standard Form)

What it is:
System with denominator order = 2 in standard form.

Key Formulas

Standard form:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Definitions:

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

Characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Damping conditions:

- ζ = 0 → Undamped
- 0 < ζ < 1 → Underdamped
- ζ = 1 → Critically damped
- ζ > 1 → Overdamped

Template

1. Compare denominator with standard form.
2. Extract:
   - ω_n
   - ζ

Tricks

- Underdamped ⇒ oscillatory.
- Critically damped ⇒ fastest without overshoot.

Example

Match:

$$\frac{25}{s^2 + 6s + 25}$$

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6. Step Response of Underdamped Second Order System

What it is:
Oscillatory response with decaying amplitude.

Key Formula

$$C(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)$$

Steady state error:

$$e_{ss} = 0$$

Tricks

- Envelope decays as $e^{-\zeta\omega_n t}$.

Mistakes

- Using wrong ω_d.

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7. Critically Damped Response

Key Formula

$$C(t) = 1 - e^{-\omega_n t}(1 + \omega_n t)$$

Facts

- No oscillation.
- Faster than overdamped.

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8. Overdamped Response

Facts:

- Two exponential terms.
- Slow response.
- No oscillations.

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9. Transient Specifications (Very Important)

What it is:
Performance measures from step response.

Key Formulas

Rise time:

$$t_r = \frac{\pi - \theta}{\omega_d}$$

Peak time:

$$t_p = \frac{\pi}{\omega_d}$$

Maximum overshoot:

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100\%$$

Settling time (2%):

$$t_s = \frac{4}{\zeta\omega_n}$$

Settling time (5%):

$$t_s = \frac{3}{\zeta\omega_n}$$

Delay time:

$$t_d = \frac{1 + 0.7\zeta}{\omega_n}$$

Template

1. Convert TF to standard form.
2. Find ζ, ω_n.
3. Compute:
    - ω_d

- t_p, t_r, t_s, M_p

Tricks

- Memorize:
  - $t_p = \pi/\omega_d$
  - $t_s = 4/(\zeta\omega_n)$

Mistakes

- Using ω_n instead of ω_d in t_p.

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10. Type of System

What it is:
Number of poles at origin of $G(s)H(s)$.

Key Facts

- Type 0 → no pole at origin
- Type 1 → one pole at origin
- Type 2 → two poles

Trick:
Higher type ⇒ lower steady state error but lower stability margin.

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11. Static Error Constants

Key Formulas

Position constant:

$$K_p = \lim_{s\to 0} G(s)H(s)$$

$$e_{ss} = \frac{1}{1+K_p}$$

Velocity constant:

$$K_v = \lim_{s\to 0} sG(s)H(s)$$

$$e_{ss} = \frac{1}{K_v}$$

Acceleration constant:

$$K_a = \lim_{s\to 0} s^2G(s)H(s)$$

$$e_{ss} = \frac{1}{K_a}$$

Template

1. Identify input type.
2. Use corresponding constant.
3. Take limit.

Tricks (Important Table to Memorize)

Type	Step	Ramp	Parabolic
Type 0	Finite	∞	∞
Type 1	0	Finite	∞
Type 2	0	0	Finite

Master Memory Table (Night-Before-Exam)

Topic	Key Formula	When Used
First order step	$1 - e^{-t/T}$	Step response
Second order TF	$\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$	Standard comparison
Peak time	π/ω_d	Oscillatory response
Overshoot	$e^{-\pi\zeta/\sqrt{1-\zeta^2}}$	Mp problems
Settling time	$4/(\zeta\omega_n)$	Time specs
Position constant	$K_p = \lim G(s)H(s)$	Step error
Velocity constant	$K_v = \lim sG(s)H(s)$	Ramp error
Acceleration constant	$K_a = \lim s^2G(s)H(s)$	Parabolic error

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CONTROL SYSTEMS – UNIT 4

One-Page Numerical Formula Sheet

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1. Standard Test Inputs (Laplace Forms)

Step input

$$R(s) = \frac{1}{s}$$

Ramp input

$$R(s) = \frac{1}{s^2}$$

Parabolic input

$$R(s) = \frac{1}{s^3}$$

Impulse input

$$R(s) = 1$$

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2. Final Value Theorem (Very Important)

$$\lim_{t \to \infty} x(t) = \lim_{s \to 0} sX(s)$$

Use this for:

- Steady state value
- Steady state error

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3. First Order System

Transfer function:

$$G(s) = \frac{1}{Ts + 1}$$

Step response:

$$C(t) = 1 - e^{-t/T}$$

Key facts:

- Time constant = **T**
- At $t = T \rightarrow$ 63.2% of final value
- Settles ≈ **4T**

Ramp steady state error:

$$e_{ss} = T$$

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4. Standard Second Order System

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

From denominator:

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

Compare coefficients to find:

- ω_n
- ζ

Damped frequency:

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

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5. Time Domain Specifications (Most Asked)

Peak time:

$$t_p = \frac{\pi}{\omega_d}$$

Rise time:

$$t_r = \frac{\pi - \theta}{\omega_d}$$

Where:

$$\theta = \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$

Settling time (2%):

$$t_s = \frac{4}{\zeta\omega_n}$$

Settling time (5%):

$$t_s = \frac{3}{\zeta\omega_n}$$

Maximum overshoot:

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100\%$$

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6. Steady State Error (Unity Feedback)

General:

$$E(s) = \frac{R(s)}{1 + G(s)H(s)}$$

Steady state error:

$$e_{ss} = \lim_{s\to 0} sE(s)$$

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7. Static Error Constants (Exam Favorite)

Position constant:

$$K_p = \lim_{s \to 0} G(s)H(s)$$

Step error:

$$e_{ss} = \frac{1}{1 + K_p}$$

Velocity constant:

$$K_v = \lim_{s \to 0} sG(s)H(s)$$

Ramp error:

$$e_{ss} = \frac{1}{K_v}$$

Acceleration constant:

$$K_a = \lim_{s \to 0} s^2 G(s)H(s)$$

Parabolic error:

$$e_{ss} = \frac{1}{K_a}$$

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8. Type of System Shortcut Table

Type	Step	Ramp	Parabolic
Type 0	Finite	∞	∞
Type 1	0	Finite	∞
Type 2	0	0	Finite

Shortcut to find type:
Count poles at origin in **G(s)H(s)**

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9. Matching Transfer Function Quickly (Numerical Trick)

Given:

$$\frac{K}{s^2 + as + b}$$

Compare with:

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

So:

$$\omega_n = \sqrt{b}$$

$$\zeta = \frac{a}{2\omega_n}$$

This saves time in exams.

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10. Quick Exam Strategy (What to Write First)

When question gives transfer function:

Step 1 → Convert to standard form
Step 2 → Find ω_n and ζ
Step 3 → Find ω_d
Step 4 → Use formulas

Most numericals follow this exact path.

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11. 10-Second Recognition Guide

If question mentions:

- Overshoot → need ζ
- Peak time → need ωd
- Settling time → need ζωn
- Step error → use Kp
- Ramp error → use Kv

Pole–Zero Diagram of a Transfer Function

What it is:
A **pole–zero diagram** is a plot in the **s-plane (complex plane)** showing:

- **Poles** → values of s that make the denominator zero
- **Zeros** → values of s that make the numerator zero

It visually shows **stability, transient behavior, and system response speed**.

Definition

For a transfer function:

$$G(s) = \frac{N(s)}{D(s)}$$

- Zeros = roots of $N(s) = 0$
- Poles = roots of $D(s) = 0$

How it is Drawn

1. Factor numerator and denominator.
2. Solve:
   - Numerator = 0 → zeros
   - Denominator = 0 → poles
3. Plot on complex plane:
   - Real axis = horizontal
   - Imaginary axis = vertical
4. Convention:
   - Poles = ×
   - Zeros = ○

Example

Given:

$$G(s) = \frac{s+2}{s(s+4)}$$

Zeros:

$$s = -2$$

Poles:

$$s = 0,\ s = -4$$

So diagram has:

- ○ at −2
- × at 0
- × at −4

Why It Matters (Exam Insight)

Pole locations tell:

Stability

- Poles in **left half plane** → stable
- Pole on **right half plane** → unstable
- Pole at origin → marginal / integrating behavior

Speed of response

- Poles farther left → faster response

Oscillation

- Complex poles → oscillatory response

Recognition Tricks (Fast Thinking)

If denominator is:

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

Poles are:

$$s = -\zeta\omega_n \pm j\omega_d$$

So:

- Real part → damping
- Imaginary part → oscillation frequency

Template to Solve in Exam

1. Write transfer function.
2. Find roots of numerator → zeros.
3. Find roots of denominator → poles.
4. Draw axes.
5. Mark × and ○ clearly.

That's usually all examiners expect.

Common Mistakes

- Forgetting pole at **s = 0** when denominator has s.
- Plotting poles and zeros reversed.
- Not solving quadratic roots correctly.

If you want, I can explain **how pole locations directly tell overshoot, damping, and stability in 30 seconds** — that connection makes Unit-4 much easier to visualize.